SEMINAR REPORT

A seminar Report on

AI-Based Deepfake Video Detection System

Submitted in partial fulfilment for the award of

THIRD YEAR DIPLOMA IN

COMPUTER ENGINEERING

By

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FOR THE ACADEMIC YEAR : 2026 – 2027

DEPARTMENT OF COMPUTER ENGINEERING

S.S.V.P.S's Bapusaheb Shivajirao Deore Polytechnic, Dhule

(Institute Code : 0059)

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(An Affiliated Institute of Maharashtra State Board of Technical Education)

DEPARTMET OF COMPUTER ENGINEERING

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ABSTRACT

Rapid progress in deep learning, especially Generative Adversarial Networks (GANs) and autoencoders, has made it easy to create highly realistic fake videos known as deepfakes, in which the face, expressions or speech of a person are replaced or manipulated. Although the technology has useful applications in film, education and accessibility, it is increasingly misused for spreading misinformation, identity fraud, defamation and online harassment, which weakens public trust in digital video as evidence.

This seminar report studies an AI-Based Deepfake Video Detection System that automatically analyses a video and decides whether it is real or manipulated. The system extracts frames, detects and crops faces, learns spatial features using Convolutional Neural Networks (such as XceptionNet or EfficientNet), captures temporal inconsistencies using sequence models (LSTM or transformer), and finally classifies the video with a confidence score. The report reviews twelve research works, explains deepfake creation methods and detection cues, describes public datasets (FaceForensics++, Celeb-DF, DFDC) and evaluation metrics, discusses advantages and limitations, and presents the project initiation details such as problem statement, scope, objectives, stakeholders, execution plan and resources.

Keywords: Deepfake, Deep Learning, CNN, LSTM, GAN, Face Forgery Detection, Video Forensics.

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CHAPTER 1

INTRODUCTION

Artificial Intelligence (AI) and deep learning are now used in almost every field, including healthcare, education, entertainment, banking and social media. Video has long been treated as a trustworthy form of evidence, because it is hard to fake a moving face and a matching voice. This situation has changed with the arrival of deepfakes. The word “deepfake” is formed from “deep learning” and “fake”, and it refers to synthetic media in which a person in an existing image or video is replaced with someone else’s likeness or made to say and do things they never did.

Most deepfake videos are produced with autoencoders and Generative Adversarial Networks (GANs). These models learn the facial appearance of a person from many images and then generate a convincing face that can be placed over another person’s face in a video. Free software, pre-trained models and consumer graphics cards have made this process available to people with very little technical skill, and fake videos can spread to millions of viewers within hours through social media and messaging applications.

The harmful use of deepfakes creates serious risks. Fake videos of political leaders can spread false information during elections, fake voices and faces can be used in financial fraud and impersonation, and non-consensual fake content can damage the dignity and reputation of ordinary people. Another side effect is that genuine videos can be dismissed as “fake”, which weakens trust in all digital media. Therefore, reliable methods to verify the authenticity of videos are urgently required.

The AI-Based Deepfake Video Detection System is designed to meet this need. It uses deep learning models to look for traces that are usually invisible to the human eye, such as blending boundaries around the face, unnatural eye blinking, inconsistent lighting, mismatch between lip movement and speech, and flickering between consecutive frames. Convolutional Neural Networks analyse each frame, while recurrent or transformer-based models study how the frames change over time. The final output is a label (real or fake) along with a confidence score.

The effectiveness of such a system is judged using accuracy, precision, recall, F1-score and Area Under the ROC Curve (AUC), and, most importantly, by how well it works on manipulation methods and datasets that it has not seen during training.

CHAPTER 2

LITERATURE REVIEW

1. FaceForensics++: Learning to Detect Manipulated Facial Images

Rössler et al. introduced FaceForensics++, a large benchmark containing real videos and their manipulated versions created with four methods (Deepfakes, Face2Face, FaceSwap and NeuralTextures). The study showed that CNN-based detectors such as XceptionNet perform well, but accuracy drops when videos are heavily compressed.

2. Celeb-DF: A Large-Scale Challenging Dataset for DeepFake Forensics

Celeb-DF provides high-quality deepfake videos of celebrities with fewer visible artifacts than earlier datasets. Its main message is that many existing detectors, which work well on older datasets, find these improved fakes much harder to detect.

3. The DeepFake Detection Challenge (DFDC) Dataset:

Created by Facebook (Meta) with paid actors, DFDC is one of the largest public deepfake datasets. It was released together with an open competition and showed that detecting fakes in diverse, real-world conditions is still a difficult problem.

4. MesoNet: A Compact Facial Video Forgery Detection Network

MesoNet proposes two small CNNs, Meso-4 and MesoInception-4, that analyse mid-level (mesoscopic) image properties instead of very fine noise or whole-image semantics. It is light and fast, which makes it suitable for quick screening of videos.

5. Deepfake Video Detection Using Recurrent Neural Networks:

Güera and Delp combine a CNN that extracts frame-level features with an LSTM that learns how those features change over time. The idea is that deepfake videos contain temporal inconsistencies between frames that a purely frame-by-frame model can miss.

6. Exposing DeepFake Videos By Detecting Face Warping Artifacts:

Li and Lyu observed that deepfake generation produces a face of limited resolution which must be warped to match the target face, leaving detectable artifacts. They trained CNNs to detect this resolution inconsistency between the face area and its surroundings.

7. In Ictu Oculi: Exposing AI Created Fake Videos by Detecting Eye Blinking

This work detects early deepfakes using a physiological cue: people blink at a regular rate, while many fake videos showed abnormal blinking because training images rarely contained closed eyes. A long-term recurrent CNN is used to classify the eye state over time.

8. Combining EfficientNet and Vision Transformers for Video Deepfake Detection:

Coccomini et al. combine EfficientNet features with Vision Transformers to capture both local texture and global relations in the face. Several models are combined by voting, and the approach is evaluated on DFDC and FaceForensics++.

9. Lips Don't Lie: A Generalisable and Robust Approach to Face Forgery Detection

LipForensics focuses on the mouth region and learns high-level irregularities of mouth movement using a network pre-trained for lipreading. Because it does not depend on low-level artifacts of one generator, it generalises better to unseen manipulation methods and compression.

10. Multi-attentional Deepfake Detection:

This paper treats deepfake detection as a fine-grained classification problem. A multi-attention network looks at several local facial regions at the same time, and a texture enhancement block highlights subtle artifacts, improving accuracy on benchmark datasets.

11. Thinking in Frequency (F3-Net): Face Forgery Detection by Mining Frequency-Aware Clues

F3-Net analyses the frequency domain of images in addition to normal pixel information. Frequency clues remain useful when compression hides visible artifacts, so the method is especially helpful for low-quality videos.

12. The Creation and Detection of Deepfakes: A Survey

Mirsky and Lee provide a broad survey of how deepfakes are created (reenactment, replacement, editing and synthesis) and how they can be detected. The survey also discusses open challenges such as generalisation and the continuous race between generators and detectors.

Table No 2.1: Literature Review

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
1	Large-scale benchmark dataset and detection study	XceptionNet And another CNN classifie rs; face tracking	FaceForensics ++(1,000 real videos,4mani pulation methods)	Provide a benchmark to train and compare forgery detectors.	Accuracy falls on Heavily compressed videos and unseen manipulation methods.
2	High-quality deepfake dataset for benchmarkig	Improved synthesis pipeline; evaluation of existing detectors	Celeb-DF v2 (590 real, 5,639 fake videos)	Offer more realistic fakes to test generalisation of detectors.	Existing detectors Score lower on it; limited to celebrity subjects.
3	Large-scale challenge dataset	Face-swap synthesis; baseline neural-network detectors	DFDC (over 100,000 clips from 3,426 paid actors)	Create a diverse, Large dataset to drive detector research.	Needs high computing power; detectors still struggle on unseen real- world fakes.
4	Compact CNN-based detection	Meso-4 and MesoInceptio n-4 networks	Online deepfake videos and Face2Face Data collected By authors	Detect forgeries From mesoscopic Image properties using a light network.	Shallow design may Be weaker on high-quality or heavily compressed fakes.
5	Hybrid CNN + recurrent (temporal) detection	CNN feature extractor with LSTM	Collected set Of real and deepfake videos	Use frame features andtemporal inconsistency across frames.	Small training set; generalisation to other generators is not assured.
6	Artifact-based detection (face warping)	CNN models (e.g.,ResNet) on face and surrounding regions	UADFV and DeepfakeTIM IT	Detect resolution inconsistency created when faces are warped.	Newer generators May leave fewer Warping artifacts.
7	Physiologica l cue (eye blinking) detection	Long-term recurrent CNN (LRCN) on eye states	Eye-blinking video data prepared by authors	Expose fakes Through abnormal Blinking behaviour.	Modern fakes can Include natural blinking; limited to eye cues.

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
8	CNN + Vision Transformer hybrid	EfficientNet With ViT; ensemble voting	DFDC and FaceForensics ++	Combine local CNN and global transformer features for better detection.	Higher computation And training-data requirement.
9	Mouth-movement (lip) based detection	Spatio-temporal network pre-trained for lipreading	Trained on FaceForensics ++; tested on Celeb- DF, DFDC and others	Learn mouth-motion irregularities for better generalisation.	Needs a clearly Visible mouth region.
10	Attention-based fine-grained detection	Multi-attention network with texture enhancement	FaceForensics ++, Celeb-DF, DFDC	Focus on several Local facial regions to catch subtle artifacts.	More complex model; cross-dataset performance can still drop.
11	Frequency-domain detection	Frequency-aware decompositio n with two-branch network	FaceForensics ++	Use frequency clues to handle low-quality compressed fakes.	Frequency clues may weaken after resizing or other post-processing.
12	Survey of creation and detection methods	Review of GANs, encoder-decoder models and detectors	Not applicab-le (survey paper)	Summarise creation techniques, detection approaches and challenges.	Presents no new system; the field changes quickly.

CHAPTER 3

INFORMATION

3.1 Overview of Deepfake Technology:

A deepfake is a video, image or audio clip that has been generated or altered using deep learning so that it looks authentic. A deepfake video detection system is a software tool that examines a video and decides whether it is genuine or manipulated. It automates the work of a digital-forensics expert by learning, from thousands of examples, the subtle differences between real and fake faces and motion.

3.2 Deepfake Creation Techniques:

Most face-swap deepfakes use an autoencoder with one shared encoder and two decoders. The encoder compresses a face into a small set of features, and each decoder learns to rebuild the face of one person. After training, the features of person A are passed through the decoder of person B, which produces B's face with A's expressions and head pose. The generated face is then blended into the video frame.

Generative Adversarial Networks (GANs) consist of a generator that creates fake images and a discriminator that tries to identify them. Both networks are trained together, and the generator slowly learns to produce images that the discriminator can no longer separate from real ones. GANs are used to create entire synthetic faces and to improve the realism of swapped faces. Graphics-based methods such as Face2Face and FaceSwap, and neural rendering methods such as NeuralTextures, are also used for face reenactment and replacement.

3.3 Types of Deepfakes:

Deepfake videos can be grouped into the following main types:

- **Face Swap:** The face of the person in a video is replaced with the face of another person.
- **Face Reenactment:** Expressions, head movement and mouth movement of a target person are controlled by another person.
- **Lip-Sync:** The mouth region is changed to match a new or synthetic audio track.
- **Entire Face Synthesis:** A completely non-existent face is created by a GAN.
- **Audio (Voice) Cloning:** The voice of a person is imitated and is often combined with a fake video.

3.4 Need for Deepfake Detection:

The need for detection comes from the speed and scale at which fake content can cause harm. A fake video can influence an election, trigger panic, support online fraud such as impersonating an officer or a family member, or damage the reputation of an innocent person. Human viewers are also poor at recognising good deepfakes. Manual verification by experts is slow and cannot keep up with the volume of videos uploaded every minute, so automatic AI-based detection is necessary to support journalists, fact-checkers, social media platforms and investigation agencies.

3.5 Clues Used for Detecting Deepfakes:

Detectors search for clues that generation methods leave behind. The commonly used clues are:

- **Blending Artifacts:** Visible or statistical edges where the generated face is merged with the original frame.
- **Face Warping and Resolution Mismatch:** The face region has a different sharpness than its surroundings.
- **Unnatural Eye and Mouth Behaviour:** Abnormal blinking, odd teeth and irregular lip movement.
- **Lighting and Colour Inconsistency:** Shadows and skin tone not matching the scene.
- **Head-Pose Inconsistency:** The face orientation does not match the rest of the head.
- **Temporal Flicker:** Small jumps in face texture between consecutive frames.
- **Frequency Artifacts:** Unusual patterns in the frequency domain, caused by up-sampling in generator .

3.6 Working Principle:

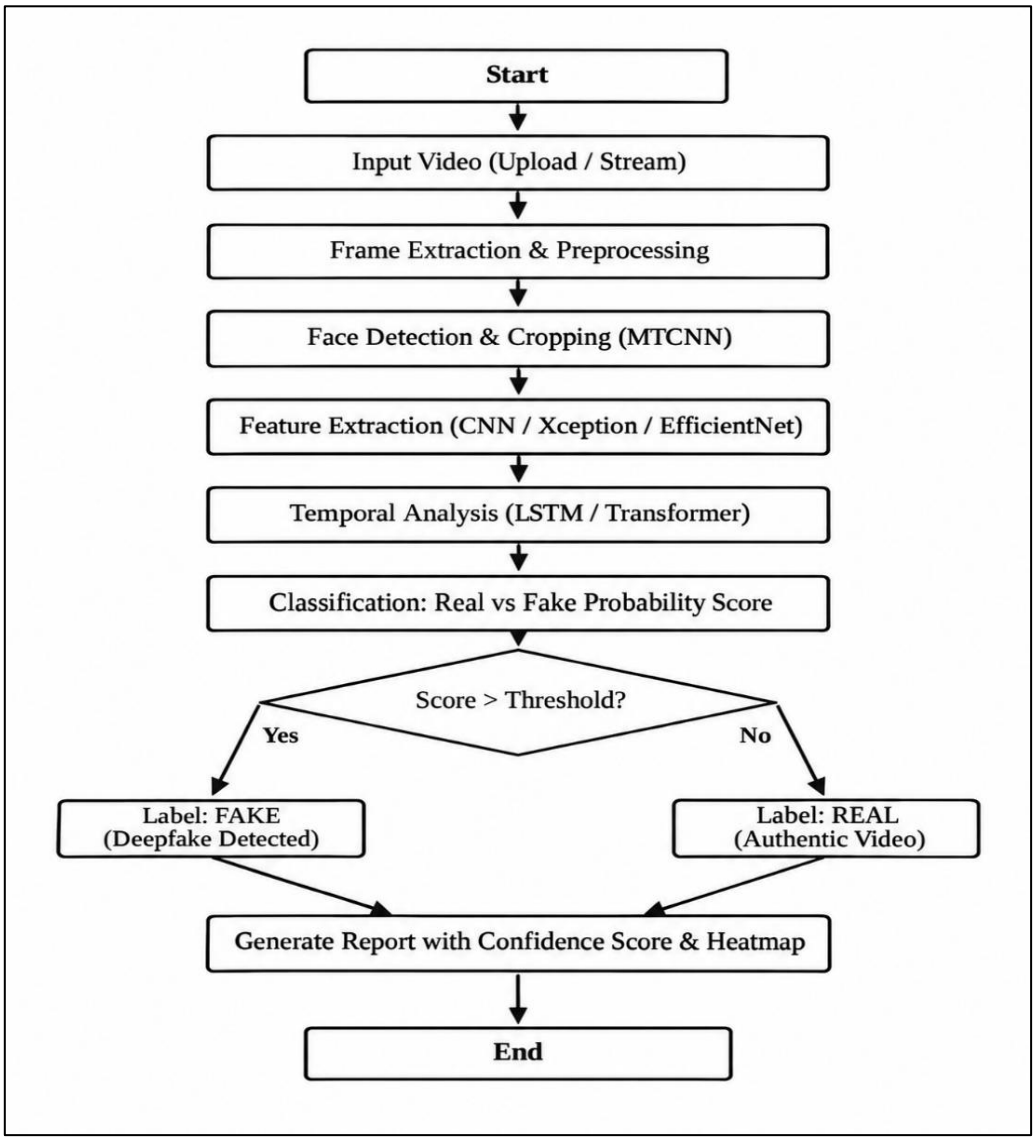


Fig 3.1: Working of AI-Based Deepfake Video Detection System

The working of the AI-Based Deepfake Video Detection System consists of the following steps:

1. **Start:** The process begins when the user wants to verify a video.
2. **Input Video:** The user uploads a video file or provides a video stream to the system.
3. **Frame Extraction:** The video is split into frames, and a fixed number of frames (for example, every tenth frame) is selected for analysis.
4. **Pre-processing:** The frames are resized, normalised and cleaned so that they match the input size required by the neural network.
5. **Face Detection and Cropping:** A face detector such as MTCNN finds the face in each frame and crops it, so that the model concentrates on the facial region.
6. **Feature Extraction:** A CNN (for example, XceptionNet or EfficientNet) converts every face image into a feature vector that captures textures, edges and blending artifacts.
7. **Temporal Analysis:** An LSTM or transformer studies the sequence of feature vectors to find inconsistencies across frames.
8. **Classification:** A fully connected layer with a sigmoid or softmax function produces the probability that the video is fake.
9. **Threshold Decision:** If the probability is greater than the chosen threshold (for example, 0.5), the video is labelled FAKE; otherwise it is labelled REAL.
10. **Result Generation:** The system prepares a report containing the label, confidence score and, optionally, a heatmap that shows the suspicious facial regions.
11. **Display Result:** The report is shown to the user through the interface.
12. **End:** The process is completed after the final result is delivered.

3.7 System Architecture and Main Components:

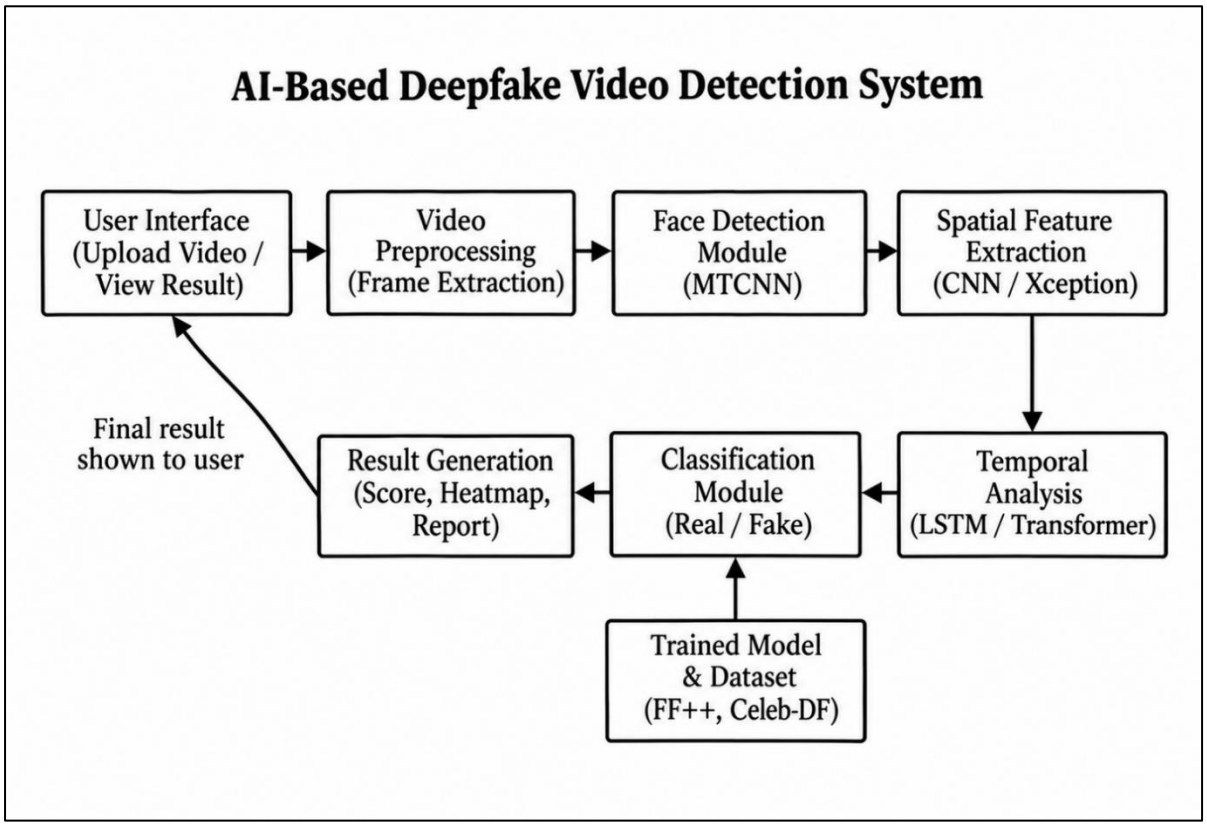


Fig 3.2: Architecture of AI-Based Deepfake Video Detection System

The main components include the User Interface, Video Pre-processing Module, Face Detection Module, Spatial Feature Extraction Module, Temporal Analysis Module, Classification Module, and Result Generation Module. A trained model and dataset store supports the classification module. These components work together to receive a video, find faces, learn visual and temporal evidence of manipulation, and present the final decision to the user.

3.8 Deep Learning Models and Techniques Used:

- **Convolutional Neural Network (CNN):** A CNN learns visual patterns such as edges, textures and blending boundaries directly from images and is the base of most frame-level detectors.
- **XceptionNet:** A CNN built on depthwise separable convolutions. It was found to be the best-performing frame-level classifier in the FaceForensics++ study and is widely used as a baseline.
- **EfficientNet:** A family of CNNs that scales depth, width and resolution together to give good accuracy at lower computation. Ensembles of EfficientNet models were popular among top solutions in the DFDC competition.
- **LSTM (Long Short-Term Memory):** A recurrent network that remembers information over a sequence of frames, so it can detect temporal inconsistencies.
- **Vision Transformer (ViT):** A transformer that splits an image into patches and uses self-attention to relate distant regions of the face; it is often combined with a CNN.

Face detection is generally performed with MTCNN (Multi-task Cascaded Convolutional Networks), which locates the face and its key points in each frame.

3.9 Datasets for Training and Testing:

Deep learning models need large labelled collections of real and fake videos. Table No 3.1 compares the commonly used public datasets.

Table No 3.1: Comparison of Deepfake Datasets

Sr. No.	Dataset	Description
1	FaceForensics++	1,000 real YouTube videos and their manipulated versions created by four methods (Deepfakes, Face2Face, FaceSwap, NeuralTextures), available in different compression levels.
2	Celeb-DF (v2)	590 real celebrity videos and 5,639 high-quality deepfake videos with fewer visible artifacts.
3	DFDC	More than 100,000 clips recorded with 3,426 paid actors, containing several face-swap methods and varied conditions.
4	UADFV	A small dataset of 49 real and 49 fake videos, used mainly in early studies on head-pose and blinking cues.

A common practice is to train on one dataset and test on another (cross-dataset testing) to measure whether the detector generalises to new kinds of fakes.

3.10 Performance Evaluation Metrics:

The performance of the detector is measured using the following metrics, where TP, TN, FP and FN are true positives, true negatives, false positives and false negatives:

- **Accuracy** = (TP + TN) / (TP + TN + FP + FN): the share of videos classified correctly.
- **Precision** = TP / (TP + FP): out of all videos flagged as fake, how many are actually fake.
- **Recall** = TP / (TP + FN): out of all fake videos, how many are caught.
- **F1-Score** = 2 × (Precision × Recall) / (Precision + Recall): the balance of precision and recall.
- **AUC-ROC:** the area under the ROC curve, which shows how well the model separates real and fake videos at all thresholds.

3.11.Example of Deepfake Detection:

Consider a 10-second video recorded at 30 frames per second, which contains 300 frames. The system selects every tenth frame, giving 30 frames. MTCNN detects the face in each selected frame and crops it to 224 × 224 pixels. The CNN converts each face into a feature vector, and the LSTM processes the 30 vectors in order.

Suppose the final layer gives a fake-probability of 0.91. Since this value is greater than the threshold of 0.5, the video is labelled FAKE with a confidence of 91%, and the heatmap highlights the mouth and jaw region where blending artifacts were found. If the probability had been 0.12, the video would be labelled REAL. (This is an illustrative example to explain the working; actual values depend on the trained model.)

CHAPTER 4

ADVANTAGES AND DISADVANTAGES

4.1 Advantages:

1. Automatic Verification: Examines videos without manual forensic effort and can screen a large number of videos quickly.
2. High Detection Capability: Deep learning models can learn subtle artifacts that human viewers cannot see.
3. Combines Spatial and Temporal Evidence: Uses both frame-level and sequence-level information, which improves reliability.
4. Supports Trust in Media: Helps journalists, platforms and investigators to verify the authenticity of video evidence.
5. Scalable and Updatable: The model can be retrained with new data as new deepfake methods appear.

4.2 Disadvantages:

1. Limited Generalisation: Detectors may perform poorly on manipulation methods or datasets not seen during training.
2. Need for Large Datasets: Training requires many labelled real and fake videos and considerable computing power.
3. Effect of Compression and Low Quality: Social media compression and low resolution can hide the artifacts the model depends on.
4. False Results: Wrongly flagging a real video or missing a fake video can cause harm to people or reduce trust in the system.
5. Continuous Arms Race: Generators keep improving and attackers can deliberately hide artifacts, so detectors must be updated regularly.

CHAPTER 5

CONCLUSION

Deepfakes created with GANs and autoencoders have become realistic enough to deceive ordinary viewers, and they pose a real threat to politics, finance, personal privacy and public trust. An AI-Based Deepfake Video Detection System provides an automatic and scalable way to verify videos by examining both the appearance of each frame and the way frames change over time.

By combining frame extraction, face detection, CNN-based feature extraction, temporal analysis and classification, the system can identify the traces left by manipulation and present the result with a confidence score. The literature reviewed in this report shows that modern detectors achieve good results on known datasets, but generalisation to new generation methods and robustness against video compression are still open challenges.

In conclusion, deepfake detection should be seen as a continuously evolving process rather than a one-time solution. With good datasets, careful evaluation, regular retraining and responsible use, such a system can become a valuable tool to protect people and organisations from the misuse of synthetic media.

CHAPTER 6

PROJECT INITIATION

6.1 Description of Problem Statement:

The rapid growth of deepfake technology has made it possible for anyone to create convincing fake videos with very little cost and skill. These videos are shared widely on social media and messaging platforms, where they can spread false information, damage reputations and support fraud before they are noticed. Human viewers cannot reliably recognise high-quality deepfakes, and manual forensic checking is too slow for the volume of content.

Existing detection methods often work well only on the datasets on which they were trained, and their accuracy drops when videos are compressed, have low resolution, or are generated with new methods. There is therefore a need for a detection system that can analyse both spatial and temporal evidence in a video and give a reliable, understandable result.

The proposed AI-Based Deepfake Video Detection System aims to address this problem by extracting frames, detecting faces, learning features with deep neural networks, analysing temporal behaviour and classifying the video as real or fake along with a confidence score.

6.2 Scope of the Project:

The scope of this project is to study and design a system that detects face-based deepfake videos. The system will accept a video, extract and pre-process frames, detect faces, analyse features using deep learning models, and display the label (real or fake) with a confidence score. A prototype will be prepared using publicly available datasets and will be evaluated with accuracy, precision, recall, F1-score and AUC. The project focuses on visual (face) manipulation in videos; audio-only deepfakes and real-time detection on live broadcasts are outside the present scope.

Objectives of the Project:

1. To study how deepfake videos are created and the artifacts they leave behind.
2. To review existing research on deepfake video detection.
3. To design a pipeline for frame extraction, face detection and pre-processing of videos.
4. To use deep learning models (CNN and LSTM/transformer) to classify a video as real or fake.
5. To evaluate the prototype using standard datasets and metrics such as accuracy, precision, recall, F1-score and AUC.
6. To present the result in a simple form, showing the label and confidence score to the user.
7. To identify limitations and suggest improvements for future work.

6.3 Stakeholders:

The stakeholders of the project are the individuals and groups who use, develop, maintain or benefit from the system.

Sr. No.	Stakeholder	Role / Interest
1	End Users / General Public	Upload videos to check whether they are real or fake, and rely on the result to avoid misinformation.
2	Social Media Platforms / Media Houses	Integrate the detector to flag fake videos and verify content authenticity before publishing or sharing.
3	Law Enforcement / Legal Authorities	Use detection reports as supporting evidence in investigations of fraud, defamation and identity misuse.
4	Government / Regulatory Bodies	Use the system for policy making, election integrity, national security and control of synthetic media.
5	Banks / Corporates / Victims of Deepfakes	Prevent fraud such as fake video KYC and impersonation, and protect personal identity and reputation
6	Developers (Project Team)	Design, train, test, deploy and maintain the AI model and the application.
7	Project Guide and Evaluators	Faculty members who guide and evaluate the project

6.4 Platform / Equipment / Resources:

1. Platform:

The system can be developed on the Windows or Linux operating system using the Python programming language. Deep learning libraries such as PyTorch or TensorFlow/Keras are used to build and train the models, and OpenCV is used for video and frame handling. Development and testing can be done in Visual Studio Code, Jupyter Notebook or Google Colab, which also provides free GPU access for training.

2. Equipment:

Table No : 6.2 Equipment

Sr. No.	Equipment	Purpose
1	Computer or Laptop	Development, model training, running the detection server and the web application (minimum 16 GB RAM recommended).
2	GPU (NVIDIA, minimum 6 GB VRAM)	Speeds up training of the deep learning model and video frame analysis during detection.
3	SSD / External Hard Disk (minimum 512 GB)	Store video datasets (such as FaceForensics++, DFDC, Celeb-DF), extracted frames and trained model files.
4	Webcam / Camera (optional)	Capture live video for real-time deepfake detection.
5	Smartphone (optional)	Record sample videos and test the application on mobile.
6	Cloud Platform such as Google Colab (optional)	Train and test models when a local GPU is not available.
7	Wi-Fi Router / Internet Connection	Download datasets and libraries, and communication between the server and users.

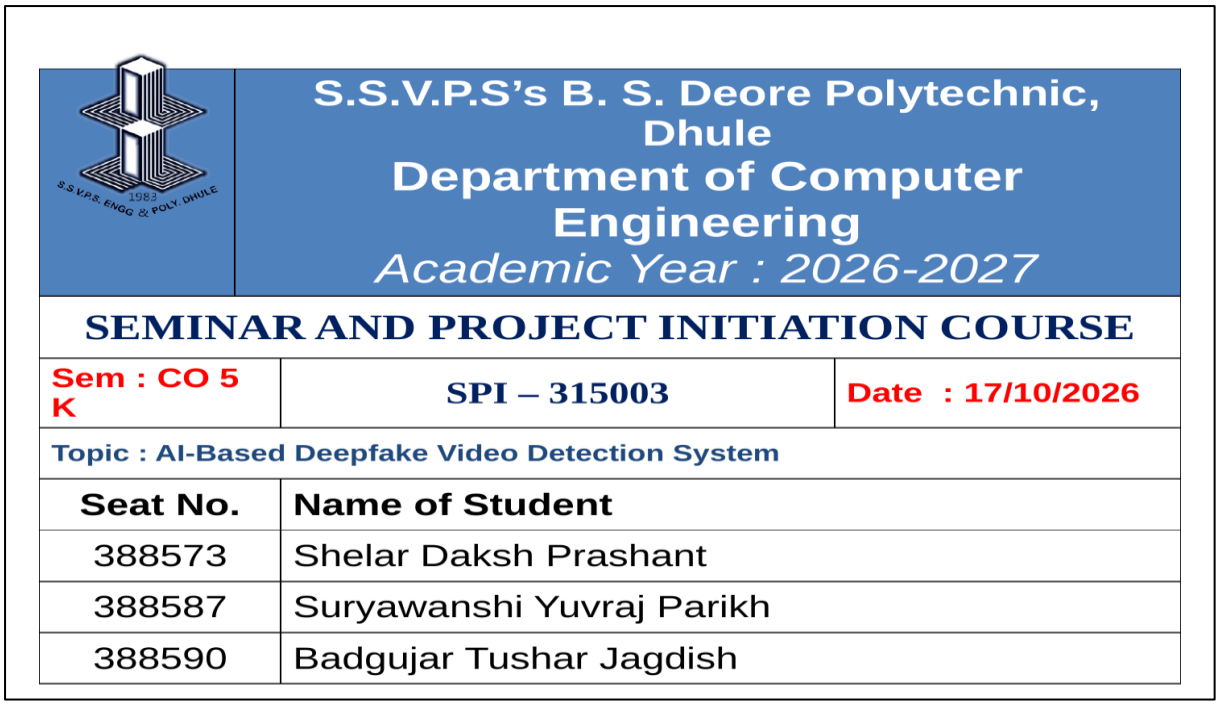
3. Resources:

Table No : 6.3 Resources

Resources	Name	Purpose
Operating System	Windows 11 / Ubuntu	Host and run the development tools and local testing environment.
Programming Language	Python 3.x	Main language for data processing, model Building and testing
Deep Learning Library	PyTorch / TensorFlow-Keras	Build, train and evaluate the CNN and LSTM/transformer models.
Computer Vision Library	OpenCV	Read videos, extract frames and perform image pre-processing.
Face Detection	MTCNN	Detect and crop faces from the extracted frames.
Data and Plotting Libraries	NumPy, Pandas, scikit-learn, Matplotlib	Handle data, calculate metrics and plot graphs.
Development Environment	VS Code / Jupyter Notebook / Google Colab	Write, run and test the code; Colab offers free GPU.
Datasets	FaceForensics++, Celeb-DF, DFDC	Publicly available real and fake videos for training and testing (use as per dataset licence).
Demo Interface (Optional)	Streamlit / Flask	Simple web page to upload a video and view the result.

CHAPTER 7

SEMINAR PRESENTATION



S.S.V.P.S's B. S. Deore Polytechnic, Dhule
Department of Computer Engineering
Academic Year : 2026-2027

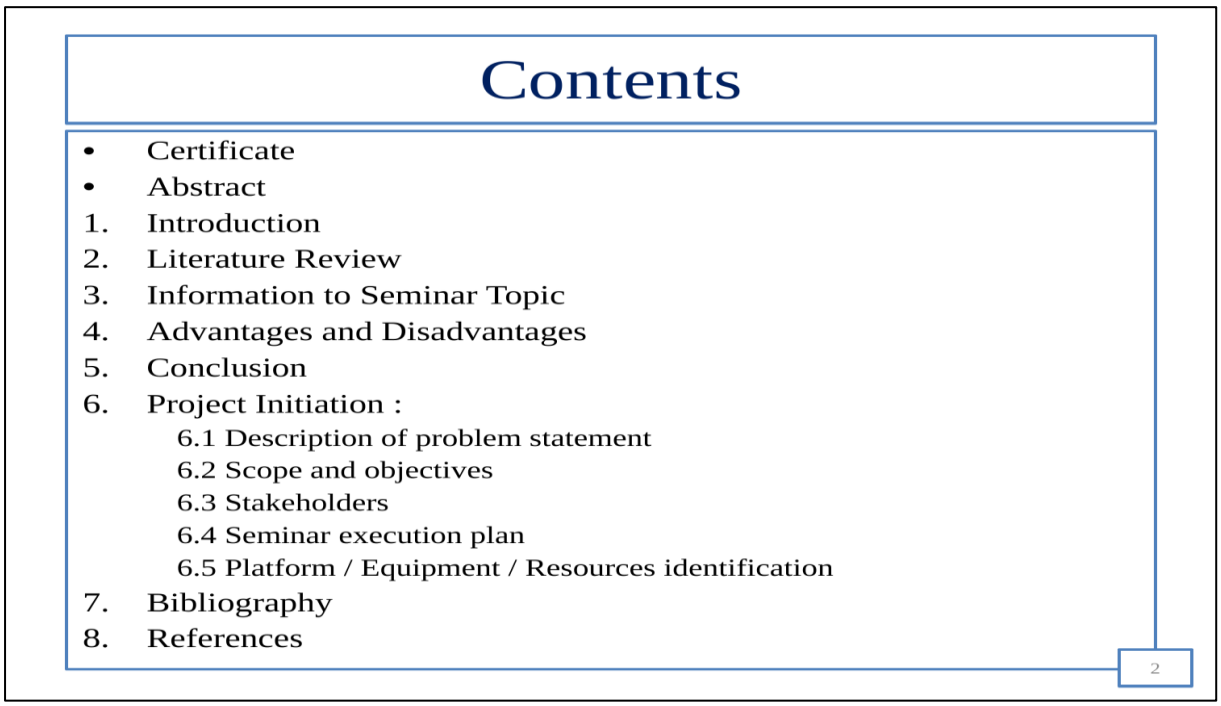
SEMINAR AND PROJECT INITIATION COURSE

Sem : CO 5 K	SPI – 315003	Date : 17/10/2026

Topic : AI-Based Deepfake Video Detection System

Seat No.	Name of Student
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388590	Badgujar Tushar Jagdish

Fig No.1: Seminar Title Page



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8. References

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Fig No.2: Content Page

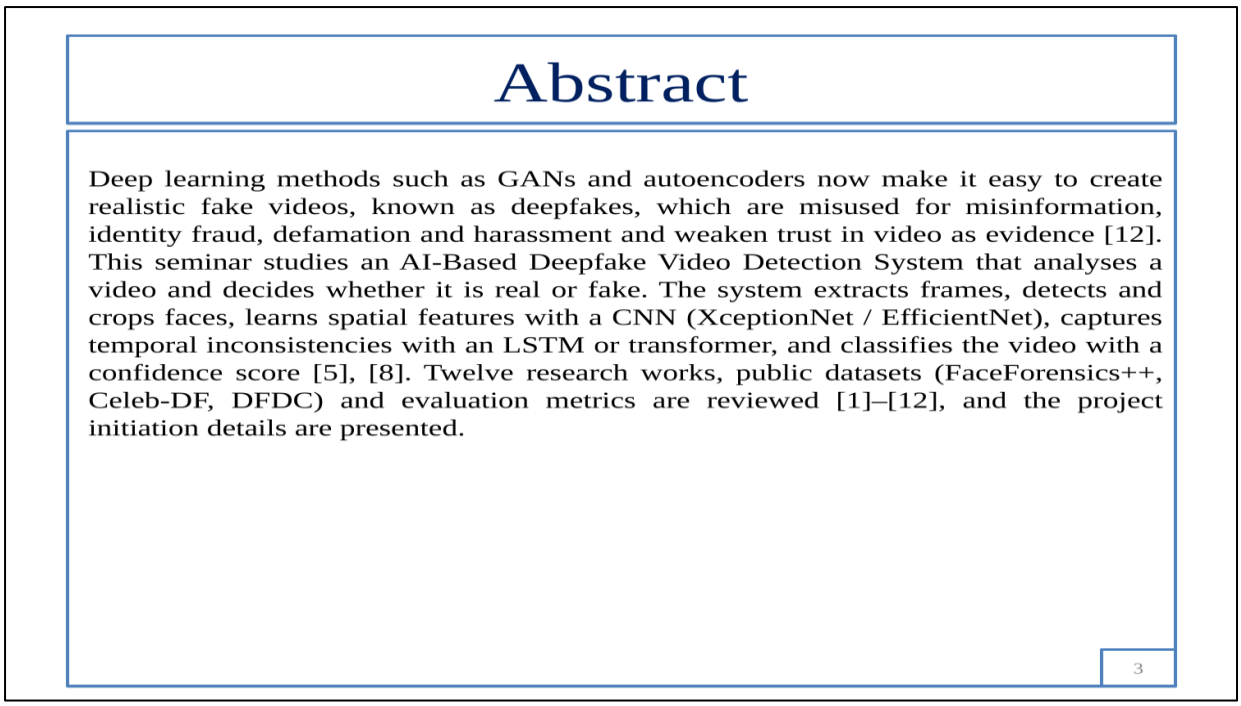


Fig No.3: Abstract Page

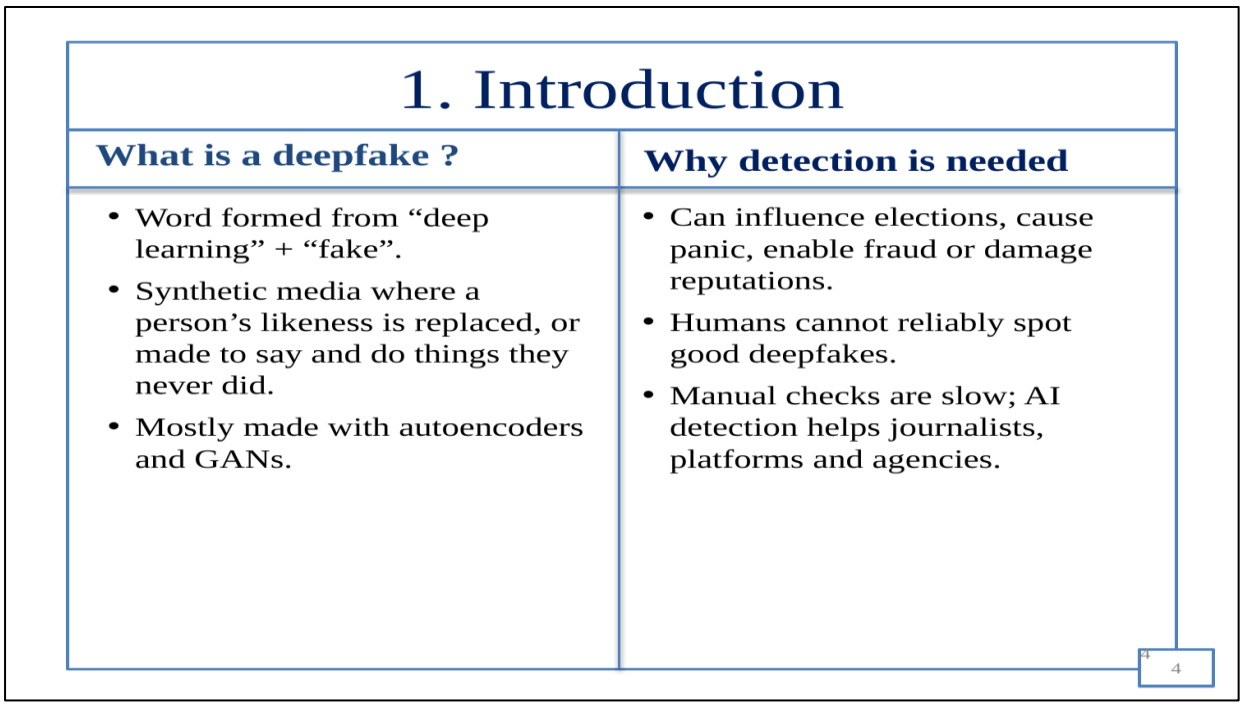


Fig No.4: Introduction Page

2. Literature Survey

Sr.	Paper and Approach Type	Technologies Used	Dataset	Key Objective	Limitations
1	**FaceForensics++ — Rössler et al. (2019)** Large-scale benchmark dataset and detection study	XceptionNet and other CNN classifiers	FaceForensics++ (1,000 real videos, 4 methods)	Benchmark to train and compare detectors.	Weak on compressed videos and unseen methods.
2	**Celeb-DF — Li et al. (2020)** High-quality deepfake dataset for benchmarking	Improved synthesis; tested existing detectors	Celeb-DF v2 (590 real, 5,639 fake)	Realistic fakes to test generalisation.	Detectors score lower; celebrity subjects only.
3	**DFDC Dataset — Dolhansky et al. (2020)** Large-scale challenge dataset	Face-swap synthesis; baseline detectors	DFDC (100,000+ clips, 3,426 actors)	Large, diverse dataset for detector research.	High computing need; struggles on unseen fakes.
4	**MesoNet — Afchar et al. (2018)** Compact CNN-based detection	Meso-4 and MesoInception-4 networks	Online deepfakes and Face2Face data	Light network using mesoscopic features.	Weaker on high-quality or compressed fakes.
5	**CNN + LSTM detector — Güera and Delp (2018)** Hybrid CNN + recurrent (temporal) detection	CNN feature extractor with LSTM	Collected real and fake videos	Use frame features and temporal inconsistency.	Small training set; generalisation unsure.
6	**Face Warping Artifacts — Li and Lyu (2019)** Artifact-based detection (face warping)	CNN models (e.g., ResNet) on face regions	UADFV and DeepfakeTIMIT	Detect resolution mismatch from face warping.	Newer generators leave fewer artifacts.

5

Sr.	Paper and Approach Type	Technologies Used	Dataset	Key Objective	Limitations
7	**In Ictu Oculi — Li, Chang and Lyu (2018)** Physiological cue (eye blinking) detection	LRCN on eye states	Eye-blinking data by authors	Expose fakes through abnormal blinking.	Modern fakes blink naturally; eye cue only.
8	**EfficientNet + ViT — Coccomini et al. (2021)** CNN + Vision Transformer hybrid	EfficientNet with ViT; ensemble voting	DFDC and FaceForensics++	Combine local CNN and global transformer features.	High computation and data requirement.
9	**Lips Don't Lie (LipForensics) — Haliassos et al. (2021)** Mouth-movement (lip) based detection	Spatio-temporal network (lipreading)	Trained on FF++; tested on Celeb-DF, DFDC	Learn mouth-motion irregularities for generalisation.	Needs a clearly visible mouth.
10	**Multi-attentional Deepfake Detection — Zhao et al. (2021)** Attention-based fine-grained detection	Multi-attention network, texture enhancement	FaceForensics++, Celeb-DF, DFDC	Focus on local facial regions for subtle artifacts.	Complex; cross-dataset drop possible.
11	**F3-Net (Thinking in Frequency) — [Authors / year: To be confirmed]** Frequency-domain detection	Frequency-aware two-branch network	FaceForensics++	Use frequency clues for compressed fakes.	Clues weaken after resizing.
12	**Creation and Detection of Deepfakes: A Survey — Mirsky and Lee (2021)** Survey of creation and detection methods	Review of GANs, encoder-decoder models, detectors	Not applicable (survey)	Summarise creation, detection and challenges.	No new system; field changes fast.

6

Fig No.5: Literature Page

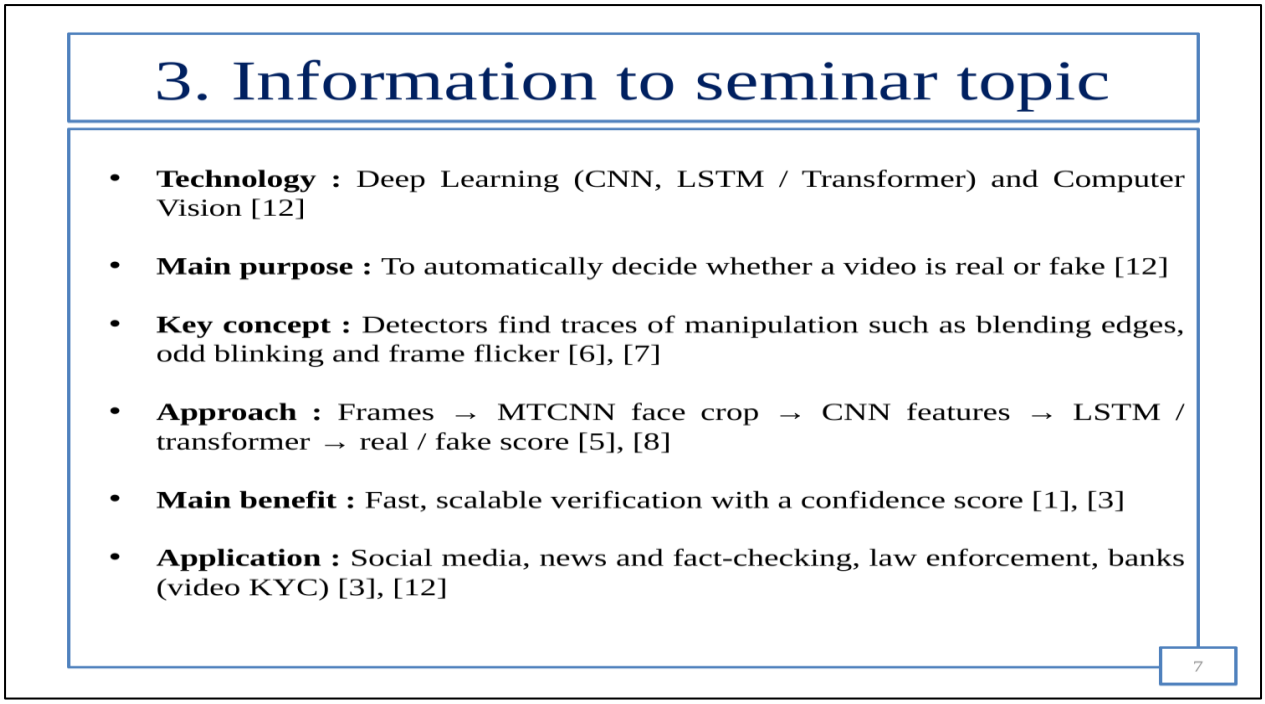


Fig No.6: Information to seminar page

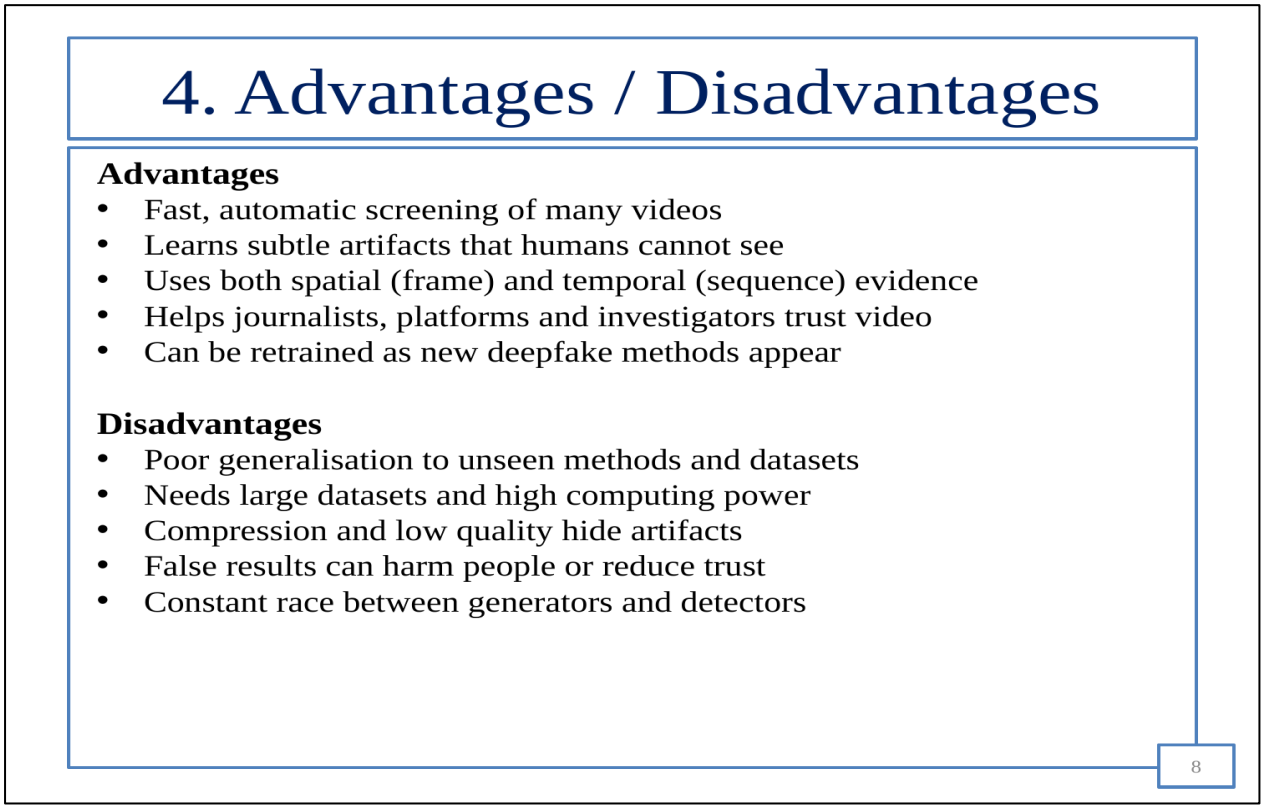


Fig No.7: Advantages & disadvantages

5. Conclusion

Deepfakes made with GANs and autoencoders are now realistic enough to deceive viewers and threaten politics, finance, privacy and public trust [12]. An AI-based detection system gives an automatic, scalable way to verify videos by studying each frame and how frames change over time [5], [8]. Modern detectors perform well on known datasets, but generalisation to new methods and robustness to compression remain open challenges [2], [9]. Detection is therefore a continuously evolving process that needs good datasets, regular retraining and responsible use.



Fig 2: Deepfake threat + AI detection (spatial + temporal) = Trusted video [5], [8], [12]

9

Fig No.8: Conclusion

6. Project Initiation

The AI-Based Deepfake Video Detection project was initiated because anyone can now create convincing fake videos at low cost, and these spread quickly on social media to cause misinformation, fraud and defamation [3], [12]. Human viewers cannot reliably spot good deepfakes and manual forensic checking is too slow [3], [12].

The project aims to build a system that extracts frames, detects faces with MTCNN, learns features with a CNN, analyses temporal behaviour with an LSTM / transformer and classifies the video as real or fake with a confidence score [5], [8].

10

Fig No.9: Project initiation

6.1 Description of Problem Statement

Deepfake videos are easy and cheap to create and spread widely on social media before they are noticed [12].

Human viewers and manual forensic checks cannot keep up with high-quality fakes [3].

Existing detectors work well only on trained datasets and fail on compressed or new fakes [2], [9]. A system using spatial and temporal evidence is needed [5], [8].

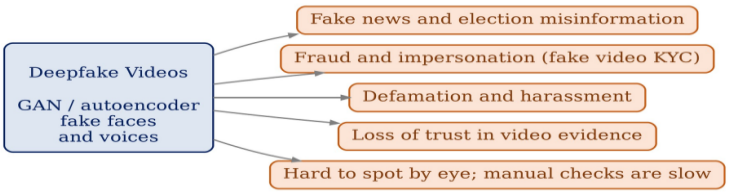


Fig 3: Problems caused by deepfake videos [3], [12]

11

Fig No.10: Description problem statement

6.2 Scope and Objectives

Objectives

- Study how deepfakes are created and the artifacts they leave [12].
- Review existing research on deepfake video detection [1]–[12].
- Build a pipeline for frame extraction, face detection and pre-processing [5].
- Classify a video as real or fake using CNN and LSTM / transformer [5], [8].
- Evaluate with accuracy, precision, recall, F1-score and AUC [1], [3].
- Show the label and confidence score in a simple form to the user.

Scope

- Detects face-based deepfake videos; prototype on public datasets [1]–[3].
- Uses Python, PyTorch / TensorFlow, OpenCV and MTCNN [5].
- Audio-only deepfakes and live-broadcast detection are out of scope.
- Helps users, platforms and investigators verify videos [12].

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Fig No.11: Scope and Objective

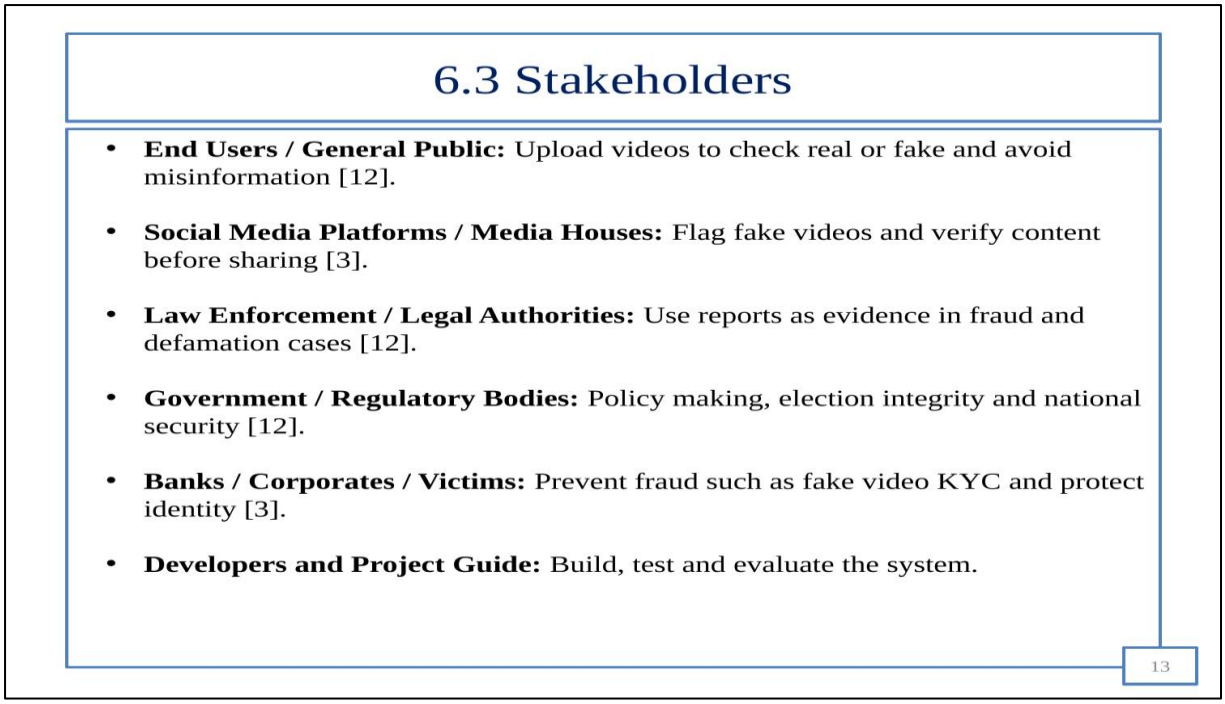


Fig No.12: Stakeholders

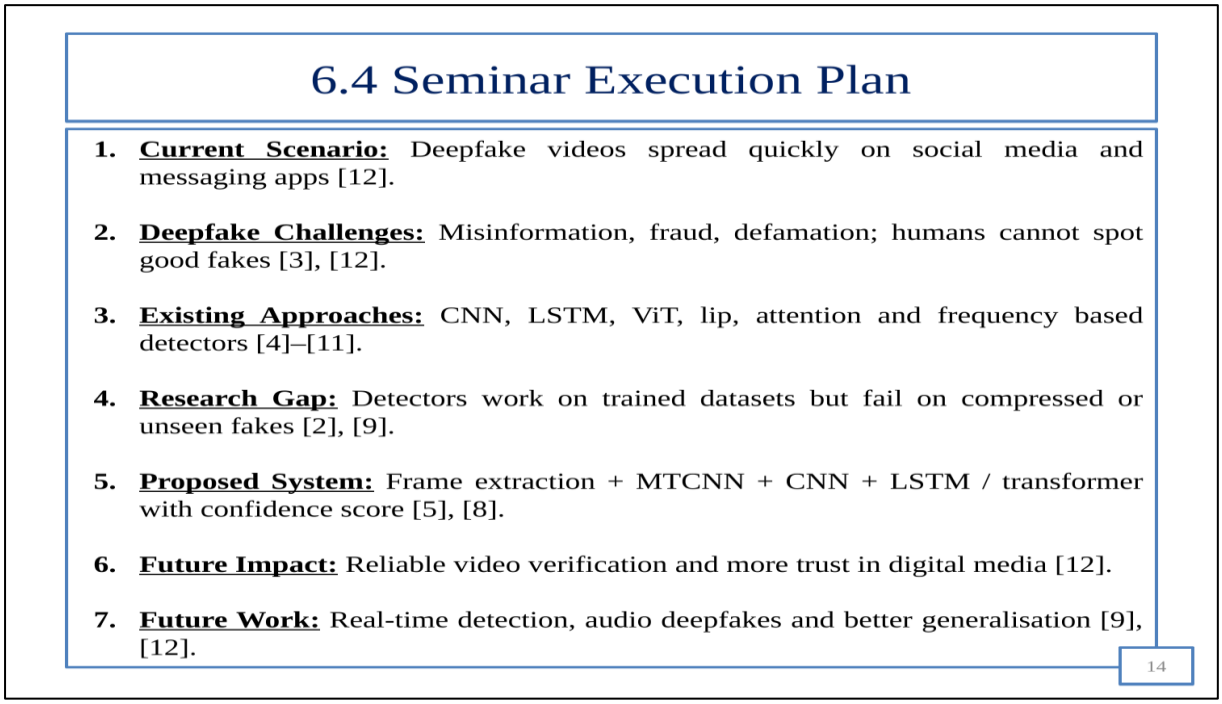


Fig No.13: Seminar execution plan

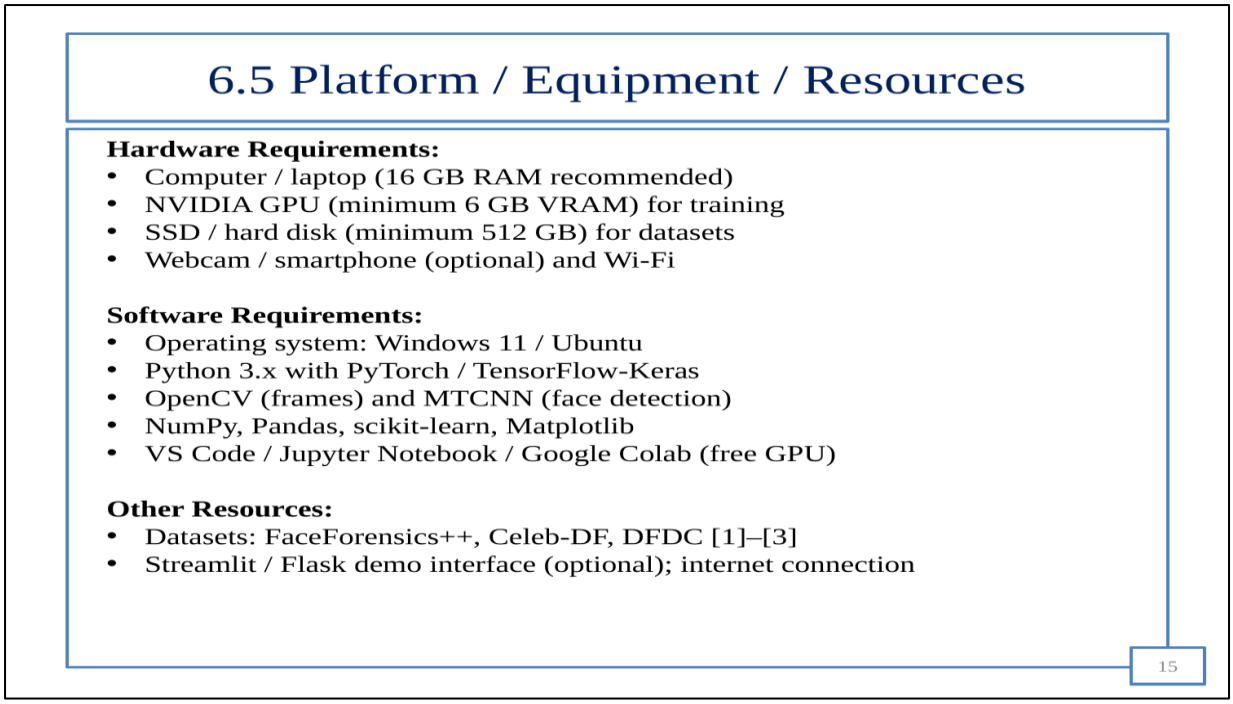


Fig No.14: Platform/Equipment/Resources

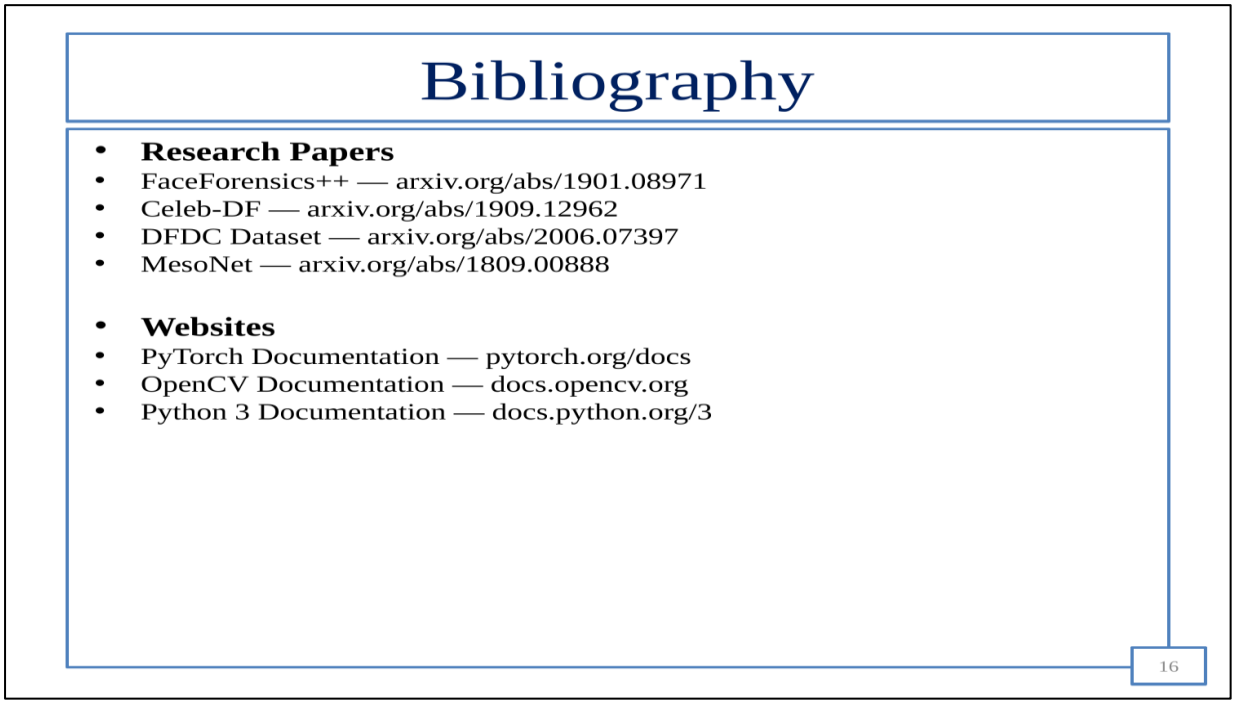


Fig No.15: Bibliography

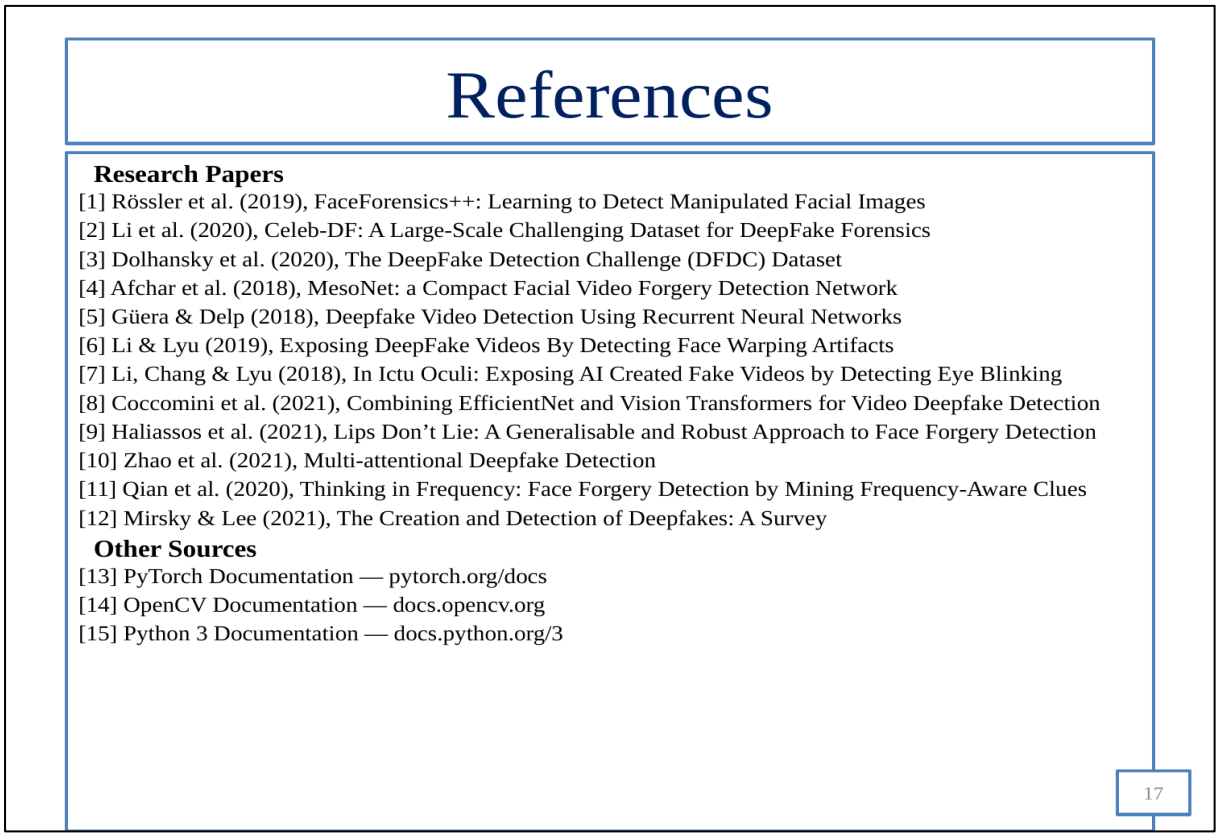


Fig No.16: References

BIBLIOGRAPHY

1. Rössler, A., Cozzolino, D., Verdoliva, L., Riess, C., Thies, J., and Nießner, M. (2019). "FaceForensics++: Learning to Detect Manipulated Facial Images." Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV). Available at: https://arxiv.org/abs/1901.08971

2. Li, Y., Yang, X., Sun, P., Qi, H., and Lyu, S. (2020). "Celeb-DF: A Large-Scale Challenging Dataset for DeepFake Forensics." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). Available at: https://arxiv.org/abs/1909.12962

3. Dolhansky, B., Bitton, J., Pflaum, B., Lu, J., Howes, R., Wang, M., and Canton Ferrer, C. (2020). "The DeepFake Detection Challenge (DFDC) Dataset." arXiv preprint arXiv:2006.07397. Available at: https://arxiv.org/abs/2006.07397

4. Afchar, D., Nozick, V., Yamagishi, J., and Echizen, I. (2018). "MesoNet: a Compact Facial Video Forgery Detection Network." IEEE International Workshop on Information Forensics and Security (WIFS). Available at: https://arxiv.org/abs/1809.00888

5. Güera, D., and Delp, E. J. (2018). "Deepfake Video Detection Using Recurrent Neural Networks." IEEE International Conference on Advanced Video and Signal Based Surveillance (AVSS). Available at: https://doi.org/10.1109/AVSS.2018.8639163

6. Li, Y., and Lyu, S. (2019). "Exposing DeepFake Videos By Detecting Face Warping Artifacts." IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW). Available at: https://arxiv.org/abs/1811.00656

7. Li, Y., Chang, M.-C., and Lyu, S. (2018). "In Ictu Oculi: Exposing AI Created Fake Videos by Detecting Eye Blinking." IEEE International Workshop on Information Forensics and Security (WIFS). Available at: https://arxiv.org/abs/1806.02877

8. Coccomini, D. A., Messina, N., Gennaro, C., and Falchi, F. (2021). "Combining EfficientNet and Vision Transformers for Video Deepfake Detection." arXiv preprint arXiv:2107.02612. Available at: https://arxiv.org/abs/2107.02612

9. Haliassos, A., Vougioukas, K., Petridis, S., and Pantic, M. (2021). "Lips Don't Lie: A Generalisable and Robust Approach to Face Forgery Detection." IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). Available at: https://arxiv.org/abs/2012.07657

REFERENCES

1. Rössler, A., et al. “FaceForensics++: Learning to Detect Manipulated Facial Images.” IEEE/CVF International Conference on Computer Vision (ICCV), 2019. https://arxiv.org/abs/1901.08971

2. Li, Y., et al. “Celeb-DF: A Large-Scale Challenging Dataset for DeepFake Forensics.” IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020. https://arxiv.org/abs/1909.12962

3. Dolhansky, B., et al. “The DeepFake Detection Challenge (DFDC) Dataset.” arXiv, 2020. https://arxiv.org/abs/2006.07397

4. Afchar, D., et al. “MesoNet: A Compact Facial Video Forgery Detection Network.” IEEE International Workshop on Information Forensics and Security (WIFS), 2018. https://arxiv.org/abs/1809.00888

5. Zhao, H., et al. “Multi-attentional Deepfake Detection.” IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2021. https://arxiv.org/abs/2103.02406

6. Mirsky, Y., and Lee, W. “The Creation and Detection of Deepfakes: A Survey.” ACM Computing Surveys, 2021. https://arxiv.org/abs/2004.11138

7. PyTorch Development Team. “PyTorch Documentation.” Official documentation for developing and training deep learning models. https://pytorch.org/docs/stable/index.html

8. OpenCV Development Team. “OpenCV Documentation.” Official documentation for image processing, video frame extraction, and computer vision operations. https://docs.opencv.org/

9. Python Software Foundation. “Python 3 Documentation.” Official documentation for Python programming, modules, and built-in functions. https://docs.python.org/3/

SEMINAR AND PROJECT INITIATION Assessment / Evaluation

Sr. No	Enrollment Number	Seat Number	Name of Student
1	24610960272	388573	Shelar Daksh Prashant
2	24610960287	388587	Suryawanshi Yuvraj Parikh
3	24610960290	388590	Badgujar Tushar Jagdish

Formative Assessment

Sr. No.	Criteria	Weightage	Exam Seat Number		
			388573	388587	388590
1	Selection Topic / Theme of seminar	5			
2	Literature review and data presentation	5			
3	Quality of Preparation and innovativeness	5			
4	Q-A handling	5			
5	Time Management	5			
6	Seminar Presentation report	10			
7	Selection of Theme of Problem Statement and its innovativeness	5			
8	Stages of development of Action plan	5			
9	Prototyping	5			
	Total	**50**			
	Scaled to 25				

Sign :

Name : .....

(Course Expert/s)